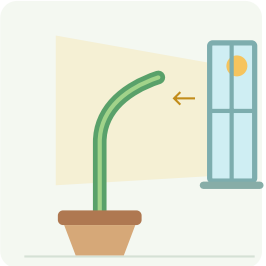
# Worksheet 3: Phototropic growth in plants

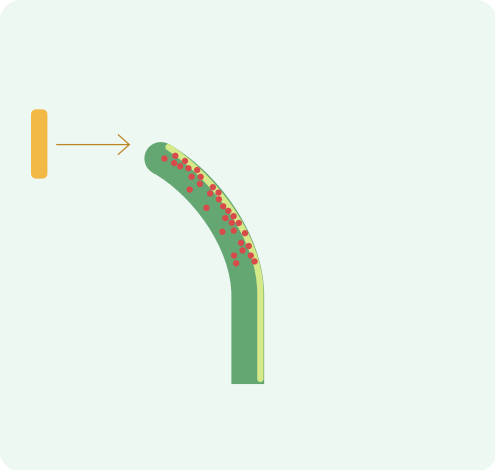
Name: _____ Class: _____ No.: _____

## Topic notes

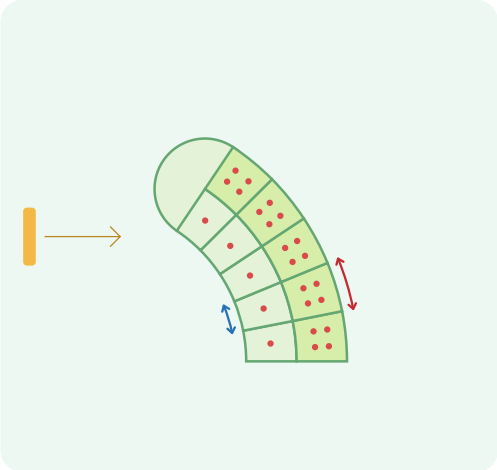


**Phototropism**（向光性） is growth in response to light direction; growth towards light is **positive phototropism**（向光性）. A coleoptile（胚芽鞘） protects young leaves.

The tip senses light. Its growth-promoting effect **passes down**. **Auxin**（生長素） is a **plant hormone**, not light or a nutrient.



Coleoptile（胚芽鞘）: more **auxin**（生長素） on the **shaded** right



Cell detail: more elongation（延長） on the right

Light on the left → **shaded** right side → more **auxin**（生長素） **passes down** → right-side cells **elongate**（延長） more → bending left.

Agar（瓊脂） supports a transferable growth-promoting effect, not its identity or distribution under light.

## Part 1 | Context and fair comparisons

Compare uncovered, tip-**shaded** and lower-**shaded** coleoptiles（胚芽鞘）. All tips are intact; other conditions stay the same.

Identify the variables: tick the appropriate factors in each row. You may choose more than one.

| Variable type | Factors to choose from |
|---|---|
| Independent variable (the factor changed on purpose) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘）type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |
| Dependent variable (the result measured or observed) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘）type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |
| Controlled variables (the factors kept the same) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘）type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |

### What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

□ The coleoptiles（胚芽鞘） start in similar growth conditions.

□ The light-blocking cap and sleeve do not limit elongation（延長） or bending.

□ The treatments do not cause clear temperature differences between setups.

□ If the plants are the same type, their starting size does not need to be controlled.

### Control group

To find out whether blocking light affects the coleoptile（胚芽鞘）'s growth direction, how would you design the control group?

(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

## Part 2 | Results and analysis

### 2.1 Shading

Light is on the left. The sleeve covers below the exposed tip without restricting growth.

A · No light blocked　　B · Light blocked at the tip　　C · Light blocked below the tip



| Setup | Bending direction |
|---|---|
| A | （Bends left ／ Bends right ／ No clear bending） |
| B | （Bends left ／ Bends right ／ No clear bending） |
| C | （Bends left ／ Bends right ／ No clear bending） |

1. Compare A and B: after **blocking light at the tip**, what changes? （No clear bending towards the light. ／ It shows the same bending towards light.）

2. Compare A and C: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend? （It can still bend towards light. ／ No clear bending towards the light.）

3. Which part may **sense light from one side**? （Tip ／ Only the lower part）

### 2.2 Tip removal

A · No light blocked　　D · Tip removed



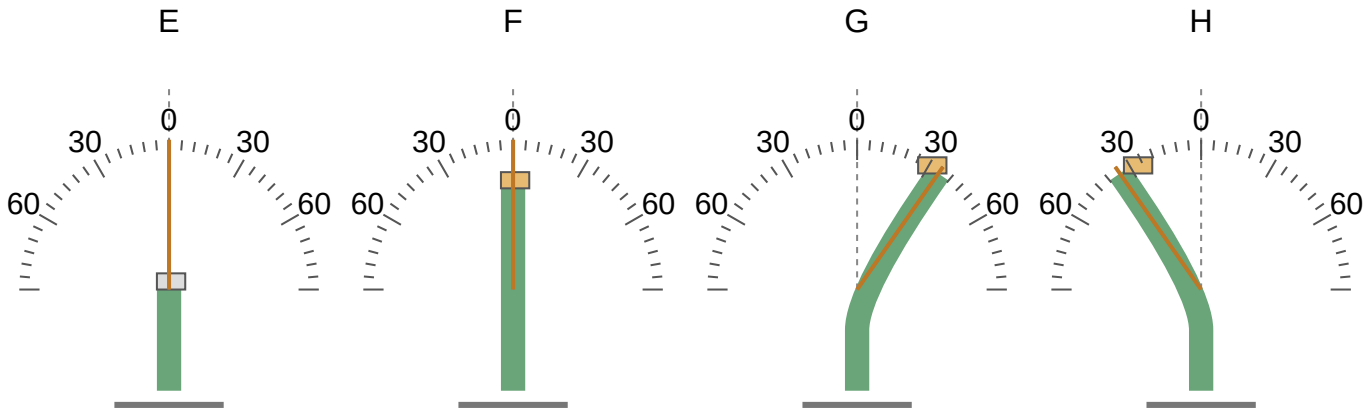
| Setup | Elongation（延長） |
|---|---|
| A | （Yes ／ No） |
| D | （Yes ／ No） |

1. What does comparing elongation（延長） in A and D support? （The presence of the tip also affects the coleoptile（胚芽鞘）'s growth / elongation（延長）. ／ The presence of the tip does not affect the coleoptile（胚芽鞘）'s growth / elongation（延長）.）

### 2.3 Agar（瓊脂）

Blank agar（瓊脂） has not touched a tip; treated agar（瓊脂） has. All cut-tip coleoptiles（胚芽鞘） stay in darkness; initial size, agar（瓊脂） size, time, temperature and water supply are the same.

E: blank agar（瓊脂）, centre; F: treated agar（瓊脂）, centre; G: treated agar（瓊脂）, left; H: treated agar（瓊脂）, right.




All start upright. Dashed lines mark vertical 0°; each small interval is 5°. E shows no clear elongation（延長）; F, G and H **elongate**（延長）. Angles and incubation are a teaching model.

| Setup | Angle (°) | Bending direction |
|---|---|---|
| E | _____ | （Bends left ／ Bends right ／ No clear bending） |
| F | _____ | （Bends left ／ Bends right ／ No clear bending） |
| G | _____ | （Bends left ／ Bends right ／ No clear bending） |
| H | _____ | （Bends left ／ Bends right ／ No clear bending） |

1. What does the comparison of E and F support? （The tip may produce a growth-promoting effect that can be transferred. ／ All agar（瓊脂） produces the same growth-promoting effect.）

2. What does the comparison of F, G and H support? （The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending. ／ The position of treated agar（瓊脂） does not affect the direction of coleoptile（胚芽鞘） bending.）

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.

## Part 3 | Integrated explanation

Use the experimental results and topic notes to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.

